

PRODUCT DESIGN & MANAGEMENT CASE STUDY

Slug Housing

Turning a UCSC housing crisis into a validated product bet

Product Strategy

User Research

Prioritization

Roadmapping

Usability Validation

KPI Definition

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Product opportunity framing · research synthesis · roadmap & metrics definition

The Opportunity, the Users, and the Product Bet



What is Slug Housing?

A student-focused housing discovery product that demystifies off-campus search near UC Santa Cruz: preference-based matching, tailored listing feeds, transparent cost breakdowns, and direct landlord messaging in place of scattered Facebook groups and email chains.



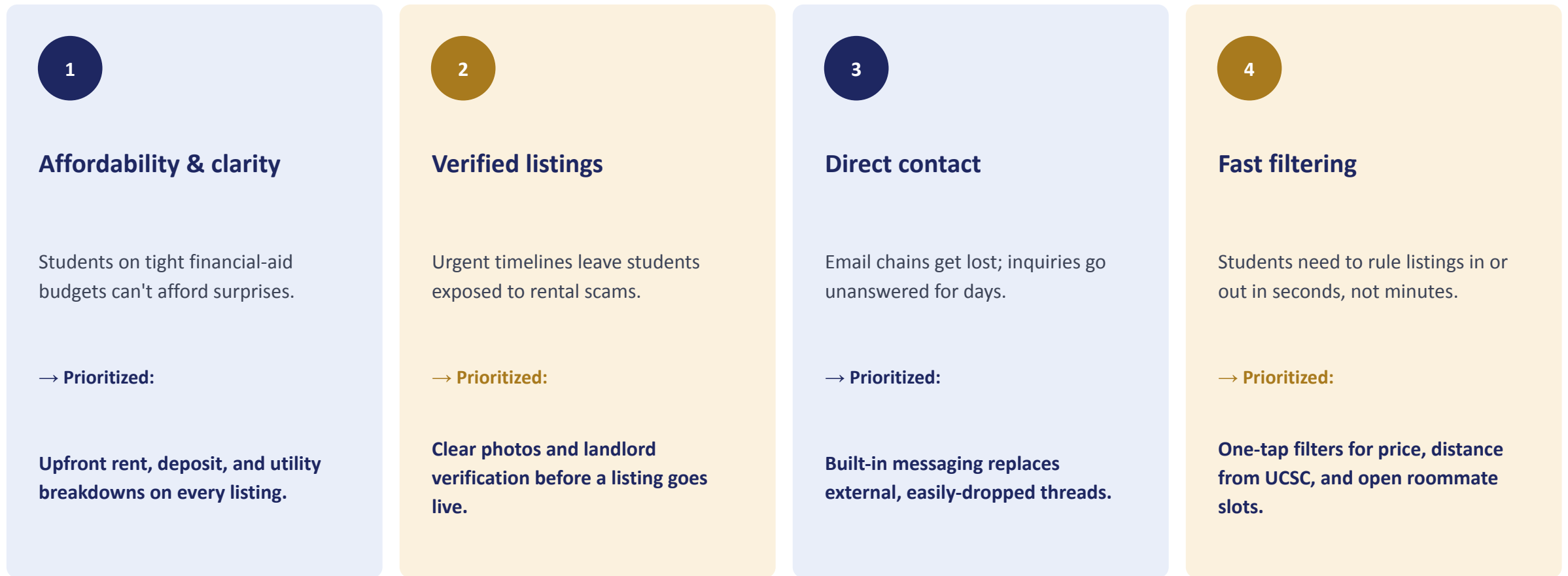
Who, and why now?

Target users are UCSC students shut out of on-campus housing by lottery cutoffs, searching under real time pressure. General real-estate platforms weren't built for them: information is fragmented, utility costs are hidden until move-in, leases rarely match a school-year timeline, and there's zero context on future housemates.

THE PRODUCT BET

If we replace open-ended browsing with a short preference survey up front, students reach listings that already fit their budget, distance, and roommate needs — cutting the anxious, unfocused scroll that dominates today's search. Every prioritization decision on the following slides traces back to testing that bet.

From Research Signal to Prioritized Requirements



Four research signals, ranked by urgency and frequency across interviews, became the requirements for our first iteration — not a wishlist.

Designing Around Two Core User Segments

PRIMARY

John Cruz — 1st-Year Student

Goal: Secure affordable housing near UCSC transit lines quickly, before the term starts.

Pain point: First time living independently, tight financial-aid budget, worried about homelessness after missing the housing lottery.

SECONDARY

Sarah Lin — Junior Transfer

Goal: Find compatible housemates and a flexible 9-month sublease near downtown Santa Cruz.

Pain point: Incompatible roommate habits in the past; struggles to evaluate utility splits and lease terms remotely.

A third lens, Marcus Vance (Santa Cruz property manager), shaped the design too: landlords are bombarded with spam inquiries and want fast, direct communication with verified UCSC applicants — the other half of why the contact flow skips email entirely.

VOICE OF THE CUSTOMER

“I’m worried about scams.” • “I don’t know which listing is reliable.” • “Is the price negotiable?” • “Can I really afford this?”

Emotional arc across the search: anxious → overwhelmed → relieved once something is confirmed. That reframed the design goal from “make browsing fun” to “get students to a confident decision fast.”

From Lottery Rejection to Signed Lease

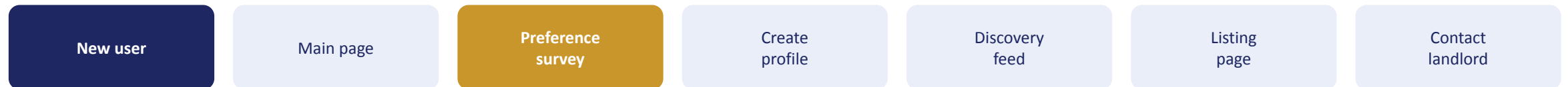
Stage	User Actions	Emotion	Expectations & Risks
1. Discovery	Loses the campus housing lottery; sees a student-targeted ad; downloads the app.	Worried / Desperate	Expects fast onboarding with no complex account barriers.
2. Exploration	Completes the preference survey; browses the UCSC housing feed.	Excited → Hopeless	Risks frustration over high prices, poor locations, or bad roommate fits.
3. Finalization	Reviews utility costs; checks housemate profiles; messages a landlord.	Hopeful	Expects a transparent cost breakdown and a clear, fast response.
4. Conclusion	Secures a lease, or keeps a filtered shortlist active as backup.	Relieved	Housing crisis resolved before the term starts.

Product implication: the sharpest drop-off is Stage 2, browsing without direction. That single risk is the argument for finishing the preference survey before any listing is shown — and the KPI most worth instrumenting first (see the metrics framework later in this deck).

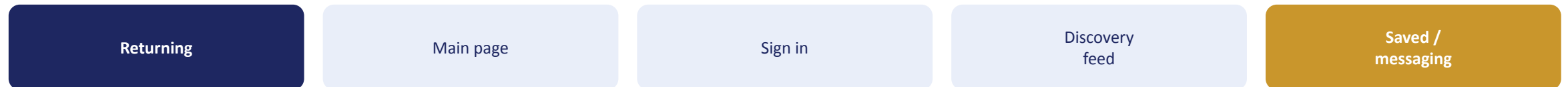
Two Onboarding Paths: First-Time vs. Returning Users

First-time users are routed through the preference survey before they ever see a listing — a direct response to the Stage 2 exploration risk. Returning users skip straight back to their saved shortlist and open conversations.

FIRST-TIME USER



RETURNING USER



WHY THIS SPLIT MATTERS

Forcing every session through onboarding would punish the students most likely to come back daily during an active search. Splitting the flow lets first-time users get the guardrails of the survey while returning users get straight back to the messages and listings they already care about — shaving repeat friction without giving up the upfront matching Stage 2 needed.

Structuring the Product Around Three Jobs to Be Done

Entry & onboarding

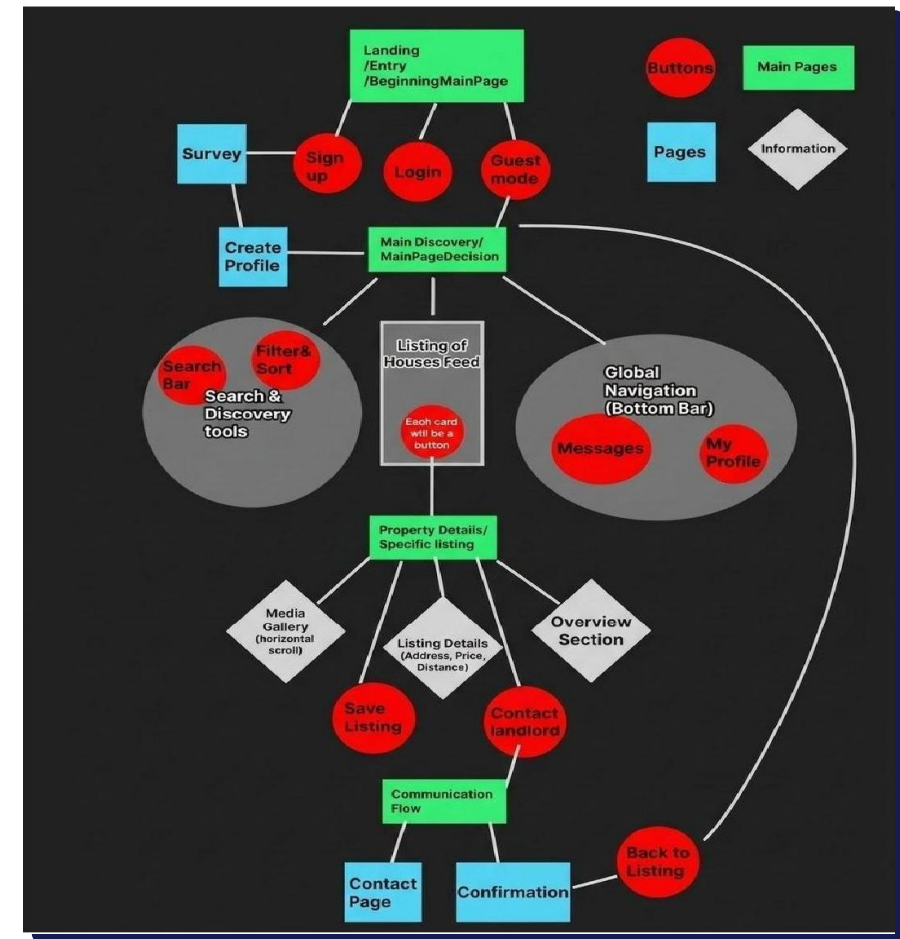
Landing page, login, sign-up, guest mode. Preference survey captures budget, distance to UCSC, and roommate count; feeds directly into profile creation.

Search & discovery

Search bar plus filter/sort controls over a card-based feed that leads with price, distance from UCSC, and a thumbnail — the three facts students scan for first.

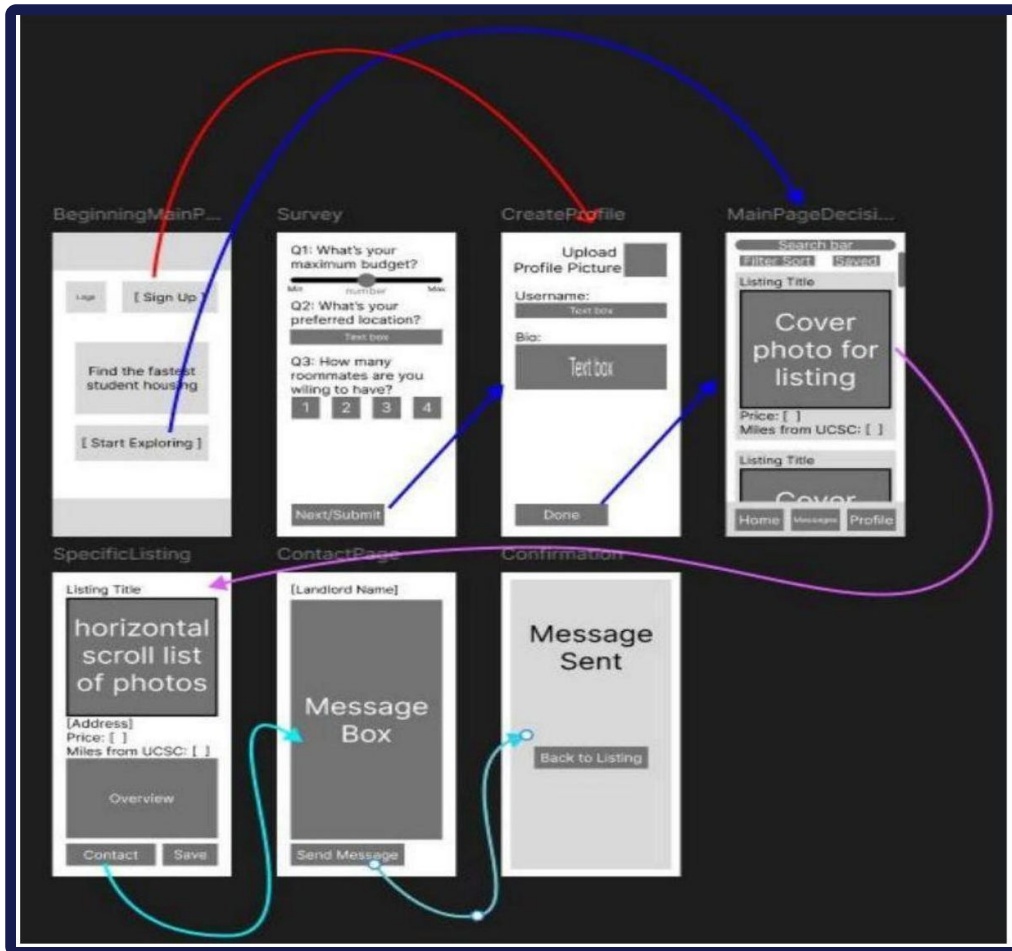
Listing & contact

Property detail combines a photo carousel, address, price, and utility overview with two actions: save the listing, or message the landlord directly.



Full site map, from Figma

De-Risking the Flow Before Higher-Fidelity Investment



Screenshot from Figma lo-fi mockups

Why low-fidelity first

Single-question screens (one budget slider, one location field, one roommate count) kept cognitive load low during an already stressful search — testable before a single high-fidelity screen was built.

What it de-risked

The messaging loop (form → visible “Message Sent” confirmation → link back to listing) was tested here first, cheaply — and it's the exact flow usability testing later flagged as needing the most rework.

Design Principles for a Stressed, Time-Pressed User

Consistent visual language

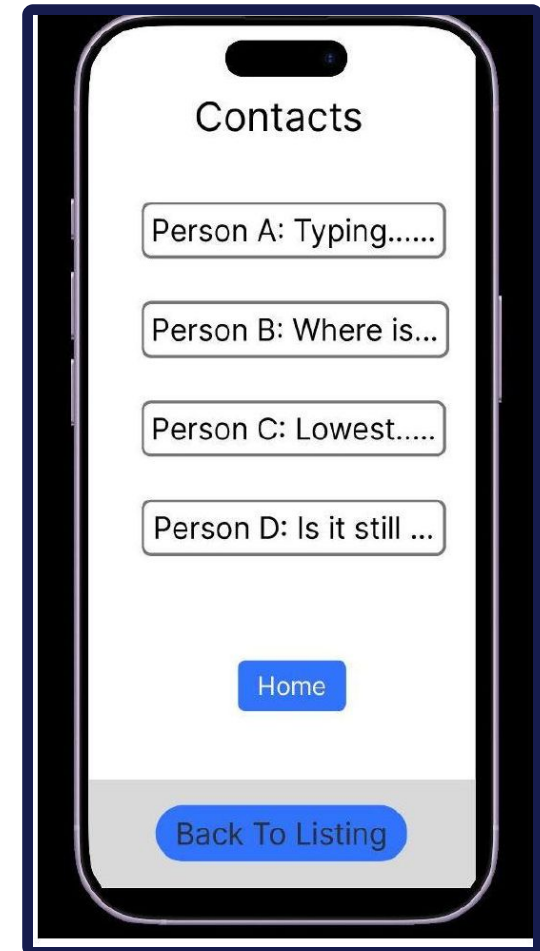
Standardized layouts, button states, and property cards so students instantly recognize what's interactive across every screen. No re-learning the UI on each page.

Calm, not cluttered

Clean interfaces deliberately reduce cognitive strain for students already stressed about budget and deadlines — a direct response to the FEEL quadrant of the empathy map.

Data over decoration

Monthly rent, distance from UCSC, and utility breakdown are visually prioritized over filler graphics, keeping attention on the facts that actually drive the decision.



Messages screen, working prototype

Usability Testing: Validating the Bet Before Scaling It

4 participants (college-aged, female), moderated think-aloud sessions — 2 remote, 2 in-person — evaluating onboarding logic, navigation, visual hierarchy, and overall intuitiveness.

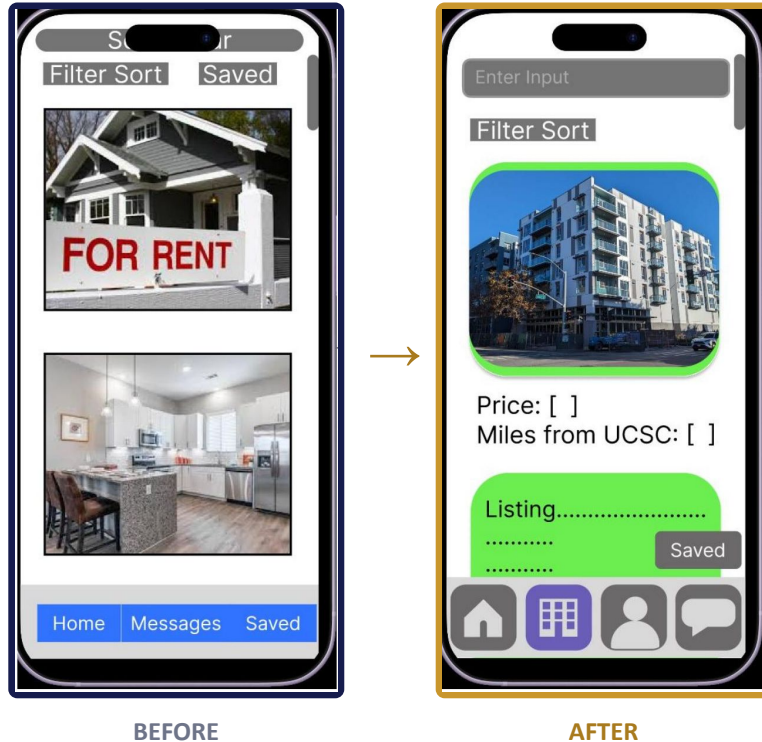
Focus Area	What We Observed	What We Changed
Layout & padding	Testers wanted more breathing room between the screen edge and body text.	Standardized top margins and a global grid across every view.
Navigation logic	Persistent bottom navigation was essential, with a clear signal for the current page.	Added active-state indicators to the global bottom navigation bar.
UI signifiers	Testers wanted clearer affordances for expandable cards and minimal icons.	Refined clickable signifiers and icon active states.
Brand & trust	High-fidelity images and consistent stylization directly influenced listing trust.	Standardized property image aspect ratios and photo stylization.

Metrics: 3 of 4 testers lost their sense of place in the app — missing/inconsistent bottom navigation, no back button on key screens, and unclear routing after onboarding. 2 testers also mistook the homepage for a login portal rather than a housing app.

ITERATION

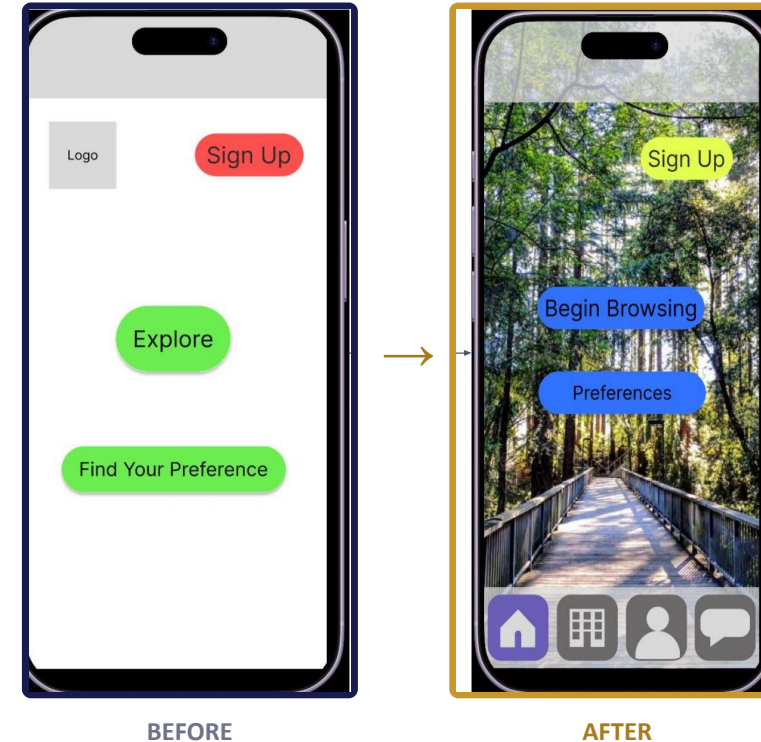
Shipping the Fix: Before & After

Bottom navigation bar



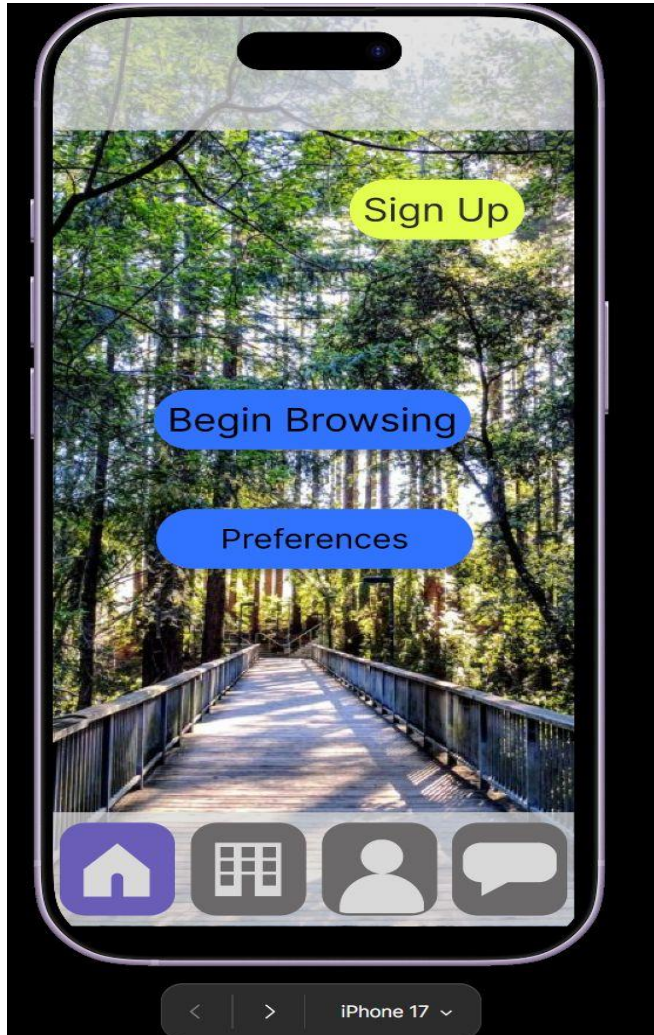
Testers lost their sense of place mid-search. Added a persistent bottom bar with a clear active-state indicator so students always know where they are.

Home / landing page



2 of 4 testers mistook the homepage for a login portal. Added imagery and clearer entry actions (Sign Up / Begin Browsing / Preferences) so the product reads as a housing app on first glance.

A Complete Survey-to-Outreach Journey



Home screen, working prototype

Intentional level of fidelity

Fidelity was calibrated for usability validation, not visual polish: the goal was to pressure-test onboarding logic, search intuitiveness, and message dispatch. The prototype covers the complete end-to-end journey from initial survey answers through final landlord outreach.

Micro-interactions & execution

Built as a fully interactive Figma prototype with attention to button click states, text input behavior, modal transitions, and navigation response (validated against real participant feedback rather than assumptions).

Try it: the interactive prototype link is included on the credits slide, alongside the initial rough-draft version for comparison.

KPI Framework: Measuring Feature Adoption & Drop-off

North Star: Search-to-lease conversion rate — % of new users who message a landlord and later mark a listing as leased / no-longer-searching.



1. Onboarding & survey

- Survey completion rate (started vs. finished)
- Time-to-complete survey
- Drop-off point within the survey



2. Discovery & filtering

- Listings viewed before first save / contact
- Feed-to-detail page click-through rate



3. Listing & contact

- Detail-page-to-contact rate
- Message send completion rate
- Time from listing view to message sent
- Landlord response rate & time



4. Retention

- Returning-user rate & days-to-return
- Saved-listing engagement rate

Priority to instrument first: survey completion rate, detail page-to-contact rate, message send completion rate, and search-to-lease conversion — these four map directly to the Exploration stage drop-off in the journey map (Stage 2) and the highest-friction flow identified in usability testing.

What This Proved, and Where It Goes Next

Student-centric solution

Bypasses generic real-estate clutter to address the specific realities of a UCSC housing crisis: lottery cutoffs, financial-aid budgets, school-year lease terms.

Preference matching

Replaces manual filtering with a short onboarding survey that matches listings to budget, proximity, and housemate fit before a student starts scrolling.

Simplified search

Rapid listing comparison, transparent utility details, and direct messaging together target the single biggest driver of student stress: time.

NEXT STEPS

- A/B test the survey-first onboarding against a browse-first flow to confirm it reduces the Stage 2 drop-off identified in journey mapping.
- Pilot landlord verification with a small set of real Santa Cruz property managers to validate the trust model beyond self-reported data.
- Instrument the prototype (or a coded build) to capture the funnel metrics defined in the KPI framework: survey completion, time-to-first-message, listing-to-lease conversion.
- Extend roommate-compatibility matching beyond a headcount field into shared preferences (cleanliness, noise, guests).

My Contribution

Sahana Vijay

- Opportunity framing & product bet definition
- User research synthesis & requirements prioritization
- Information architecture & user flow design
- Usability testing & iteration
- Figma prototyping
- Roadmap & KPI framework definition

Project Prototypes

Initial Figma Prototype

[Early concept & initial interaction draft](#)

Working Prototype

[Interactive Figma prototype execution](#)

OTHER PROJECT MEMBERS

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